

OSI Model

The seven layers of the OSI model include Application, Presentation, Session, Transport, Network, Data Link, and Physical.

An example protocol for each layer includes Application Layer – HTTP; Presentation Layer – SSL/TLS; Session Layer – NetBIOS; Transport Layer – TCP; Network Layer – IP; Data Link Layer – Ethernet; Physical Layer – transmission media, including cables.

TCP is a protocol that works at the Transport layer.

IP Addressing

Public IP is used on the internet and is globally unique, while Private IP is used on the LAN and cannot be accessed through the Internet.

Three sets of private IP addresses include 10.0.0.0 - 10.255.255.255, 172.16.0.0 - 172.31.255.255, and 192.168.0.0 - 192.168.255.255.

192.168.1.100 is a Private IP address.

Protocols

Port 22 is for SSH. Port 80 is for HTTP. Port 443 is for HTTPS. Port 3306 is for MySQL.

TCP is a Connection-Oriented Protocol and reliable, while UDP is Connection-Less Protocol and fast but unreliable.

NAT changes Private IPs to Public IP and is essential to share one Public IP among several devices and for security reasons.